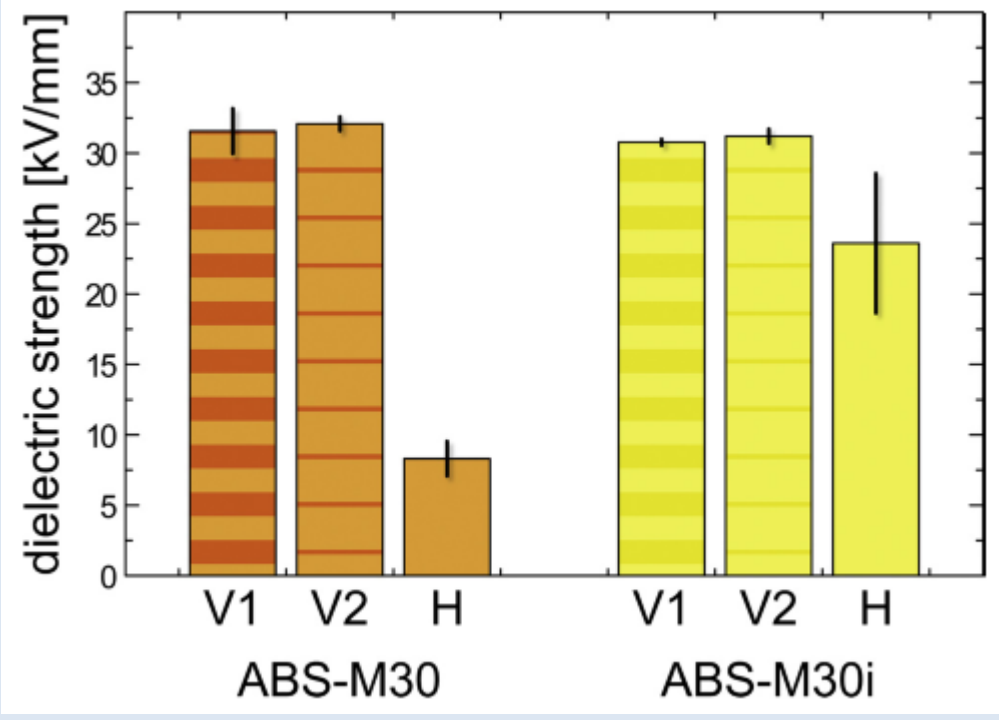


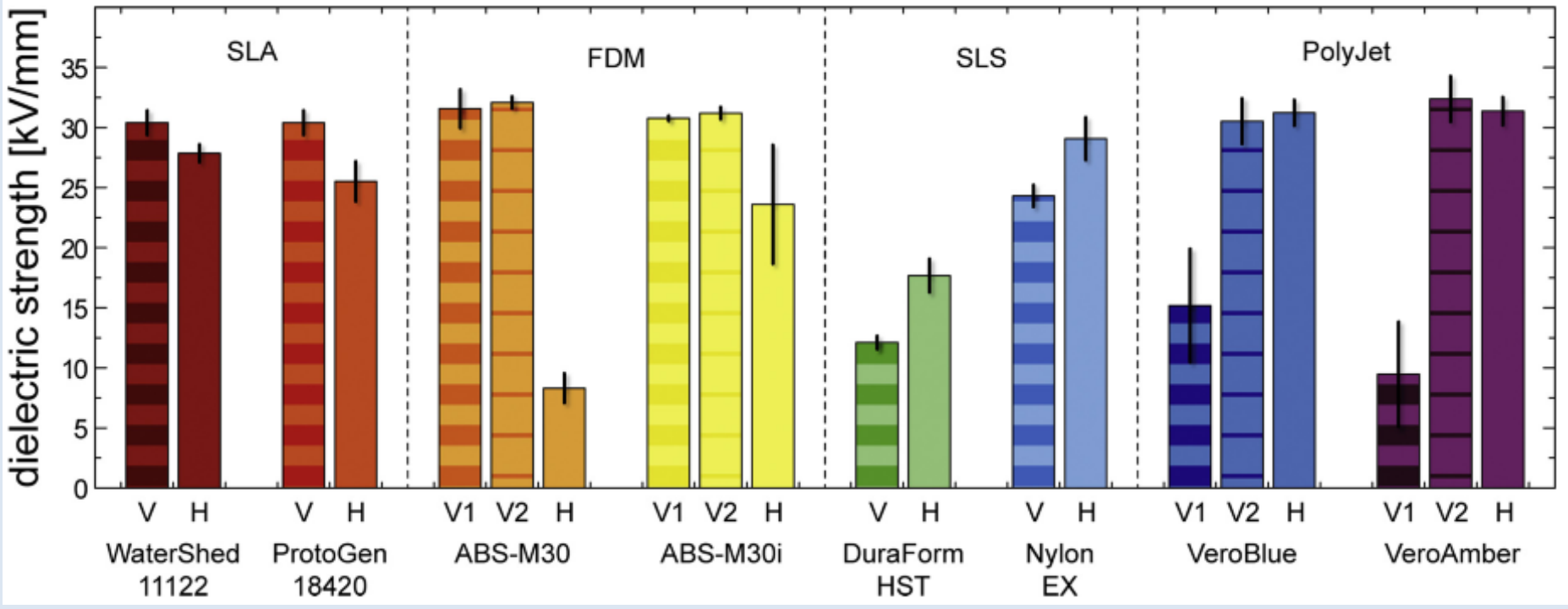
3D printed insulators

There has been much interest in utilising additive manufacturing to ‘print’ insulators – this could enable complex shaping not readily available through traditional machining and could significantly shorten lead times in producing components. To date, though, most people seeking to build high voltage parts with additive manufacturing for have relied on SLA – Stereolithography – printing, with a liquid resin cured by UV light. FDM – Fibre Deposition Modelling – in which a spool of plastic fibre is locally melted and deposited by a heated nozzle is rarely used, despite dominating the mass consumer market.

FDM printing is not typically used as tests date suggest the dielectric strength of prints made via. this method is highly dependent on the direction of printing, as highlighted by, for instance, French *et al.* (2015 IEEE Pulsed Power Conference) and Hoff *et al.* (Additive Manufacturing **22**, 2018):



Here the V1 and V2 represent printing ‘vertically’ with 100 x 100 x 1mm samples printed in the y, z, x and x, z, y direction and H means horizontally with x, y, z. Testing was performed to the ASTM D149-20 standard, with an ever-increasing voltage applied at 60hz. In both cases the ABS material has weaker dielectric strength when printed horizontally (extremely weak with ABS-M30) and often has a large variance/irreproducibility. The likely mechanism for the reduced dielectric field strength is thought to be a high degree of ‘aligned void spaces’ between the layers of the print. By comparison the results of SLA printing, alongside less utilised SLS and Polyjet methods, is shown below:



⇒ only SLA based prints appear to have relatively high, reproducible dielectric strengths in both the vertical and horizontal print directions

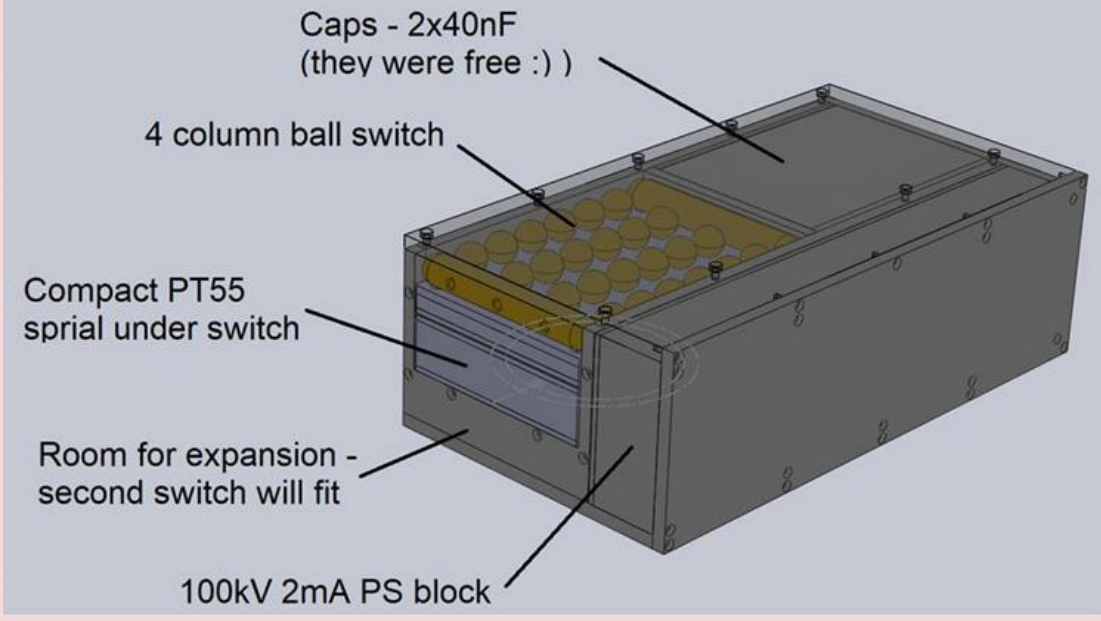
However, SLA printing is relatively expensive (>\$100/kg vs <\$40/kg for FDM), often relatively slow, and due to relying on optics to transmit the UV light build volumes for SLA printers are typically limited to ~30x30x20cm on the largest consumer units. The materials utilised by SLA printers are typically proprietary and are regularly much softer/ structurally weaker than the materials FDM can access – which includes ABS, PLA, Nylon, Pet-G, and carbon reinforced materials.

SLA printers are also limited to a single material print. A multi nozzle FDM printer can not only lay dissolvable support material (a common use) it could potentially also be used to incorporate semiconductor layers to grade material near connections – e.g. capacitor bushings - or add RF/EMI shielding in situ to a build. FDM printing is supported by the largest base of users – and, importantly, FDM printers are generally not ‘locked’ to a certain set of printing parameters. Instead FDM printers usually have manufacturers recommendations for a particular fibre (at least at the consumer end of the market) and users can further tweak temperatures, speeds and flow rates etc. to improve their prints.

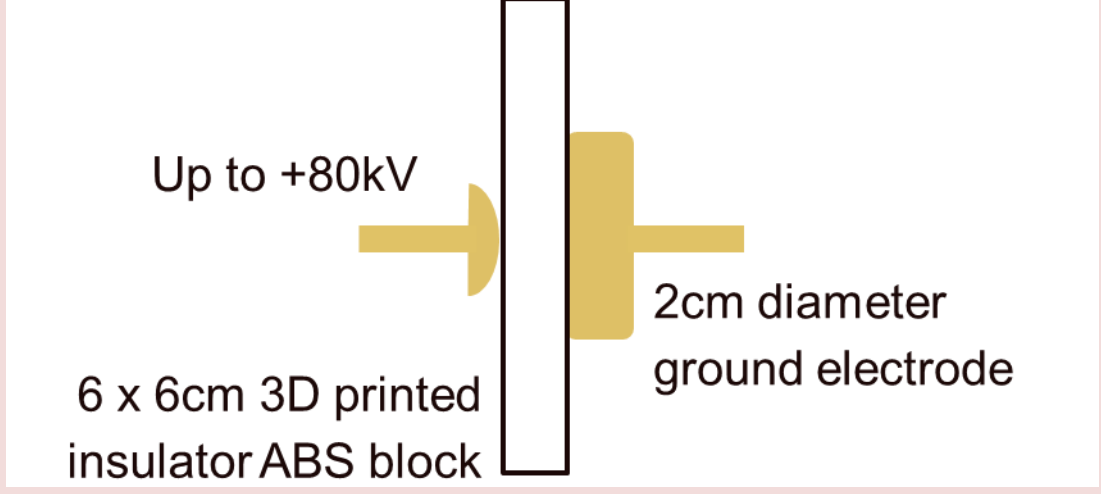
Can we adjust the parameters of consumer 3D printers to improve the dielectric field strengths of fibre deposition additive manufacturing?

Initial development – a necessity, not an experiment !

In 2018 we were developing new HV bricks for LTD generators – each was ~ 300x200x550mm and were air/plastic insulated for operation at 100kV



Making the plastic insulator/casing however proved difficult. Using screwed together sheet plastic led to breakdowns at the joins. Milling single plastic blocks was extremely costly and led to significant waste + would not enable much of the internal shaping we wished to use. Hence, we initially explored making the insulators by additive manufacturing.



Experiments were performed on a G10 frame submerged in Diala insulating oil to prevent surface flash-overs. The voltage, applied via. a Glassman power supply was increased up to +80kV. A Raise 3D pro 2 printer provided the material samples, all at 100% fill. Both horizontal and vertical prints were produced.

With ‘standard’ settings at a 100µm layer height samples printed in either direction had to be at least 10mm thick to hold off 80kV (just). This is far lower dielectric strength than cast ABS (typically 15-30kV/mm) Many samples broke, with vertical prints this was clearly seen through the print layers.



By altering the temperature, infill speed and especially the overfill settings of the printer though we found we could significantly reduce the thickness of ABS required to hold off 80kV whilst maintaining print quality and accuracy:

Method	Thickness (mm)	Hold off printed vertically (kV)	Hold off printed horizontally (kV)
Std settings, 100µm	10mm	Just 80kV	78kV
HV settings v1, 100µm	3mm	78kV	75kV
HV settings v1, 100µm	5mm	75kV	>80kV
HV settings v1, 100µm	7mm	>80kV	>80kV

3 factors that can influence the dielectric strength of additive manufactured insulators are the material, quality of the printer (rigidity, accuracy etc) and the print settings.

2 consumer level printers were utilized in the tests – a Raise 3D Pro 2 and a Bambu X1-C. The Pro 2 is approx. 6 years old (recently refurbished) and cost ~\$3500 – it was the original printer used for making HV insulators in 2018 and has been in continual use since. The Bambu X1-C is brand new and cost ~\$2000. It includes many features to help improve print quality including auto-bed levelling and Lidar for distancing the print head. Its also a factor of ~5x faster than the Pro 2.

Both printers were used to print a variety of different materials including:

- ABS - Acrylonitrile butadiene styrene –the oldest 3D printing material; a good all-rounder with high rigidity, strength and impact resistance, and good chemical resistance including oil. But shrinks significantly in printing and needs heated/hot chamber.
- PLA - Polylactic acid – a common ‘easy to print’ material; somewhat weaker than ABS, biodegradable.
- PET-G - Polyethylene terephthalate glycol – the strength of ABS, but more flexible. Easier to print with lower shrinkage. Good chemical resistance.

Both printers were used to make a set of samples 60 x 60 x 1 to 7mm thickness in 0.5mm increments. Both horizontal and vertical samples were printed.

For our experiments HV insulators are generally used in 2 situations. In the first they are used to insulate capacitors / components being charged up to high voltage over a few seconds before being discharged. In the second case they are used to isolate protect short pulses (~100ns or less) – e.g. in MITLs or the voltage pulses from trigger systems. Hence rather than rely on ASTM D149-20, which is expensive to perform commercially and at 60Hz does not really represent our use, we developed independent tests in for DC hold off – with experiments in oil over several seconds charging to 70kV – and for pulse hold off – in air with ~50ns FWHM sinusoidal signals of ~70kV.

Printing and testing



Raise 3D Pro 2, build volume 305x305x300mm – Pro2 plus expands this to x605 vertically

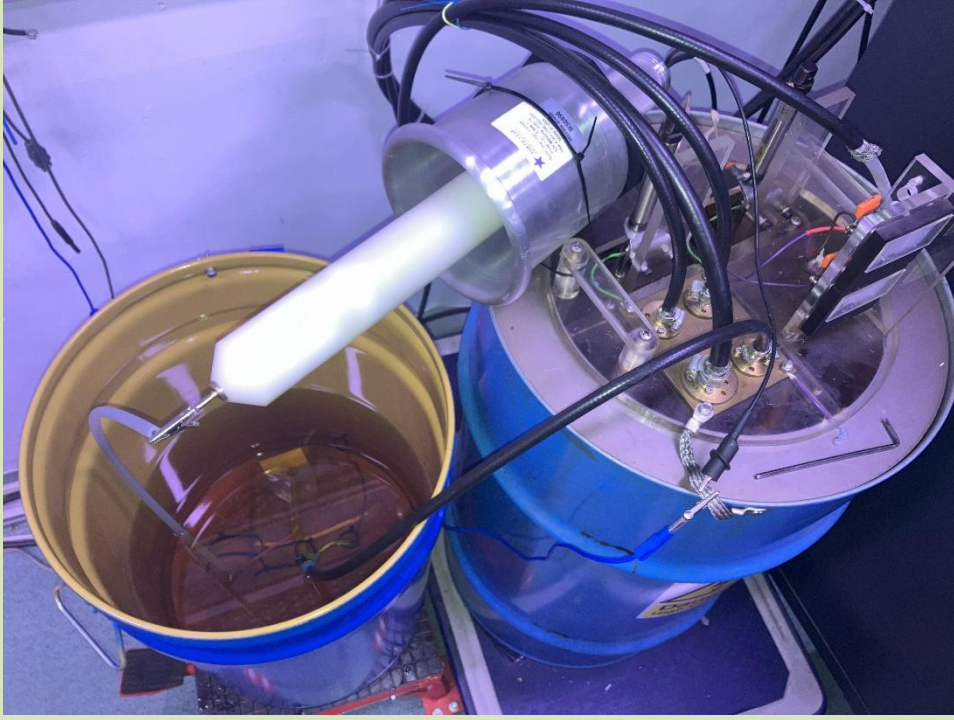
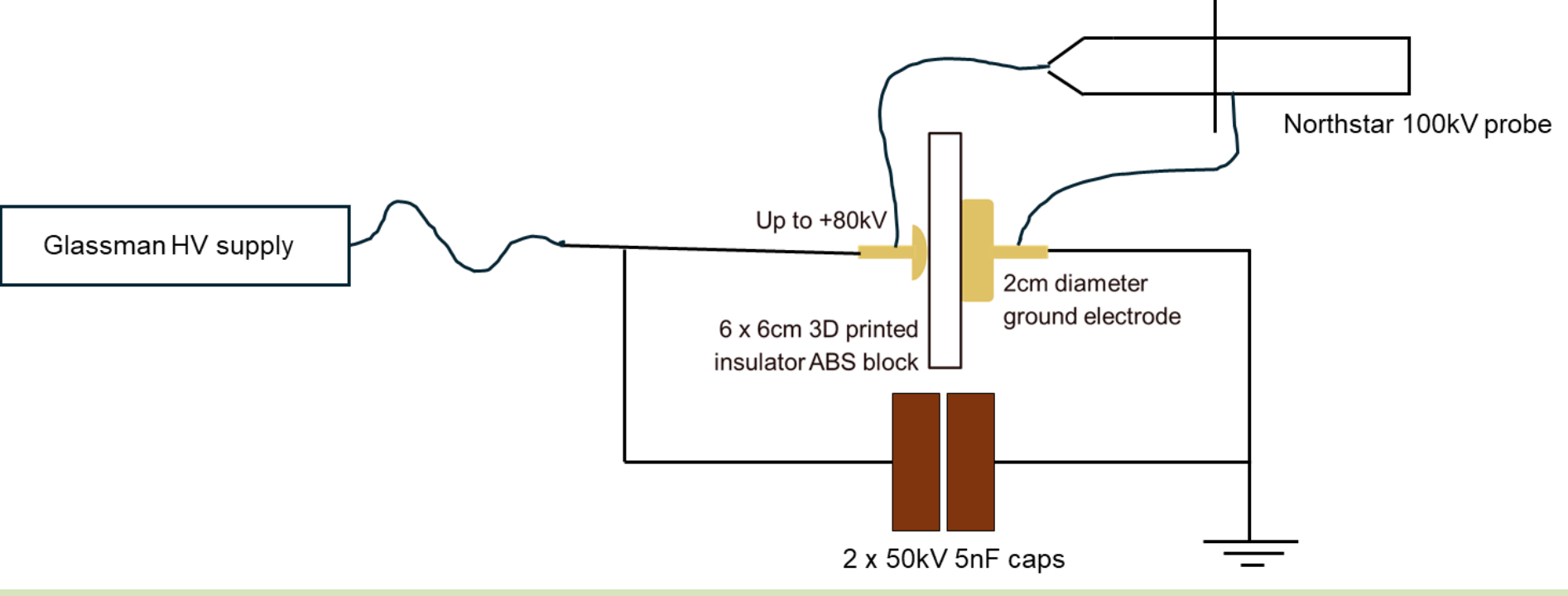


Bambu X1-C, build volume 256x256x256mm

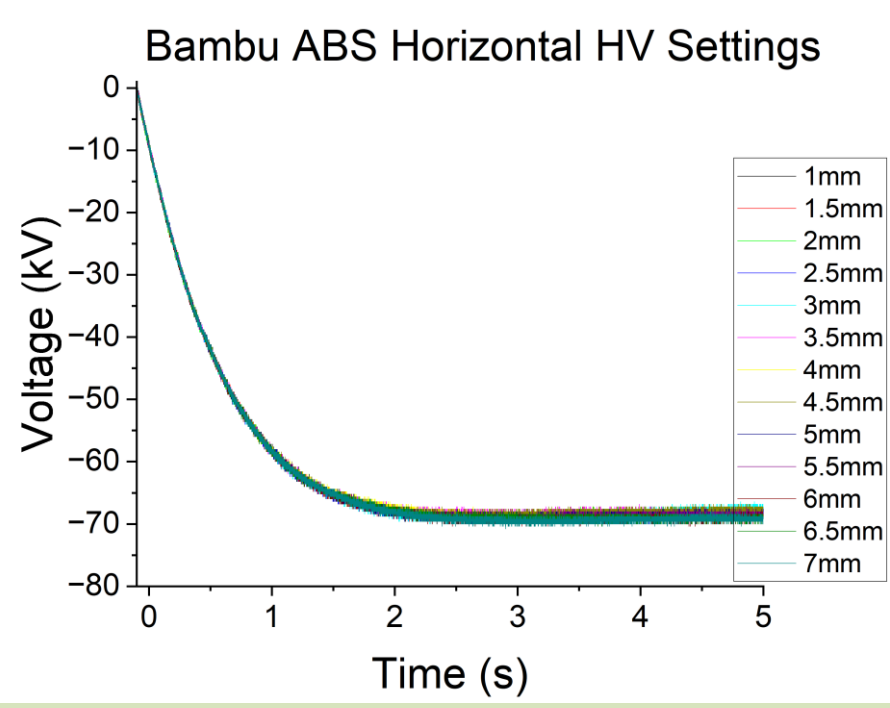
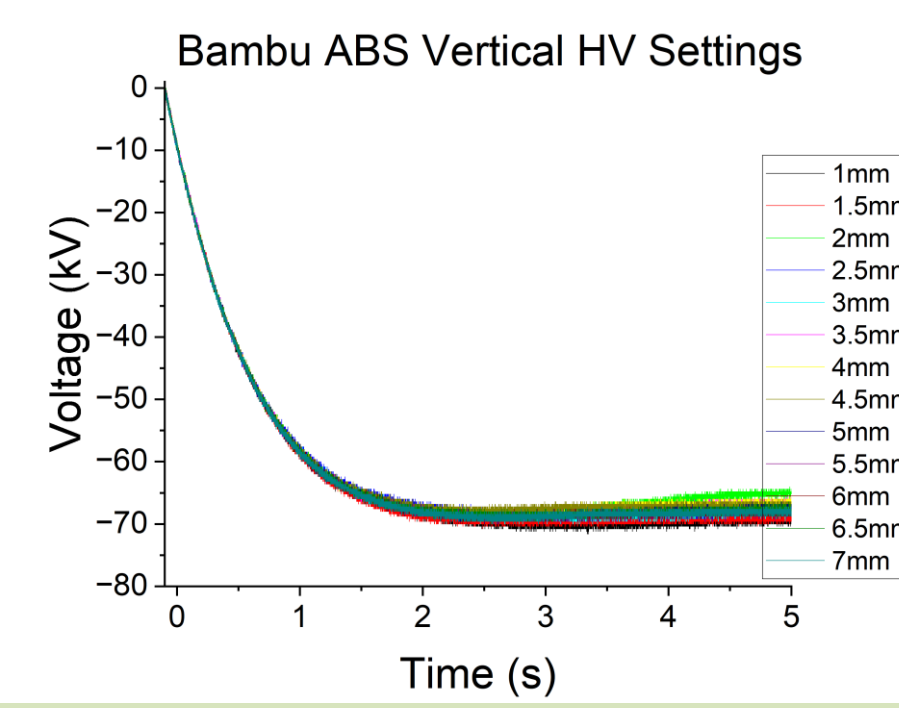
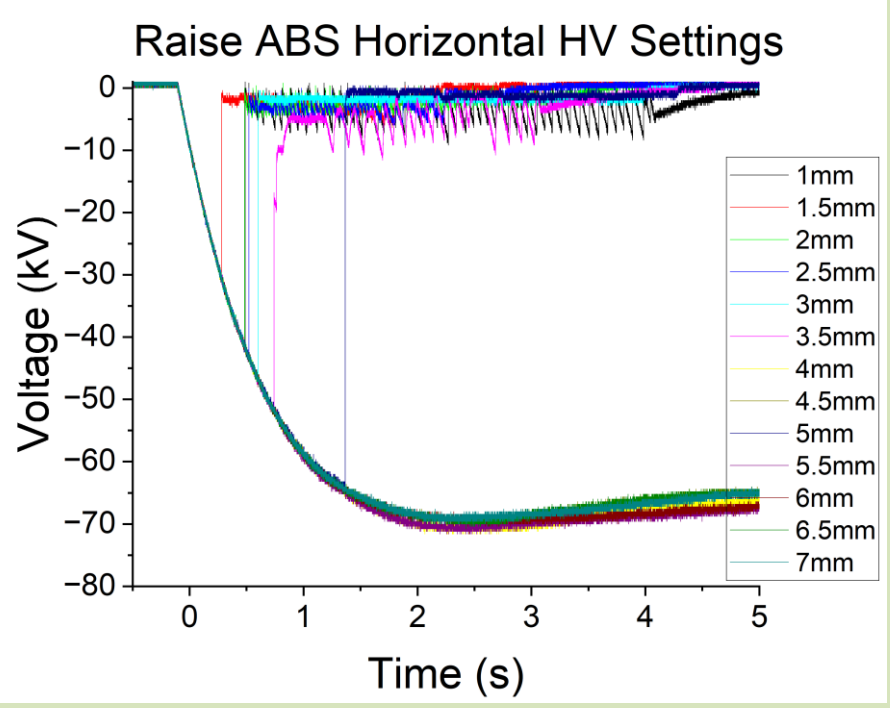
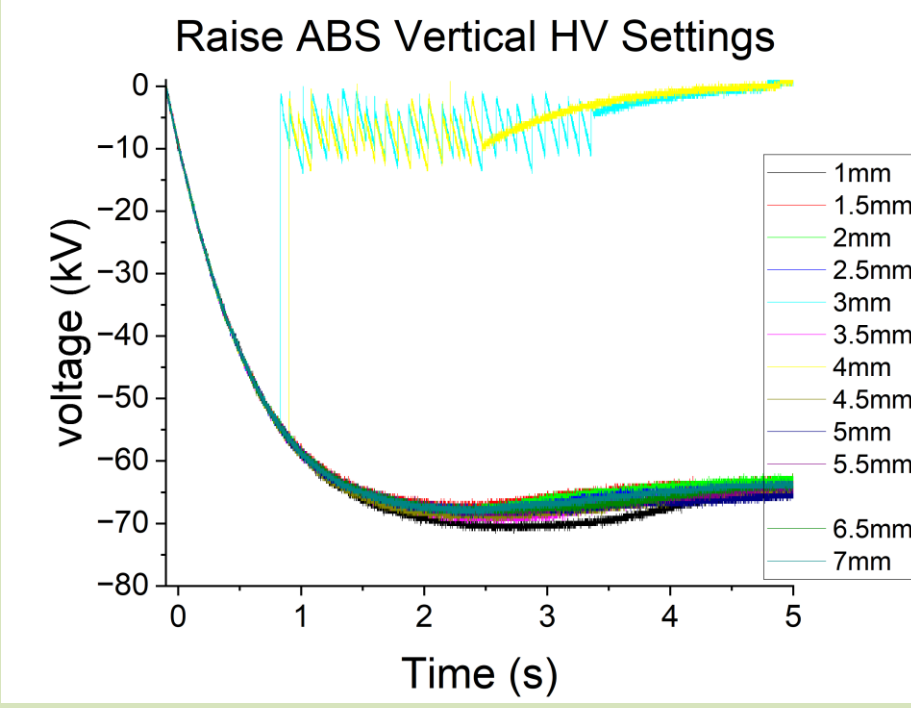
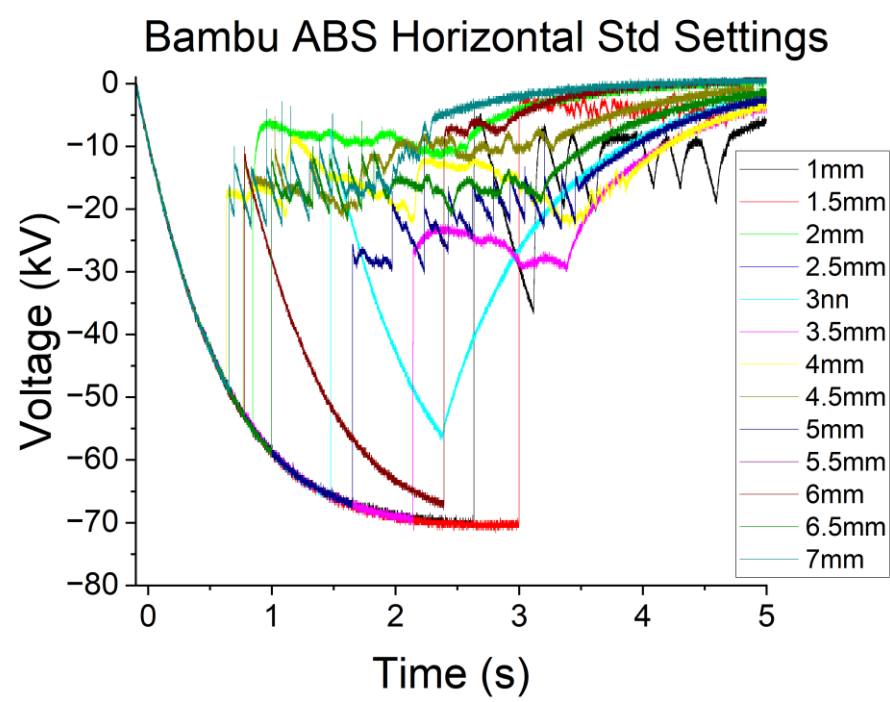
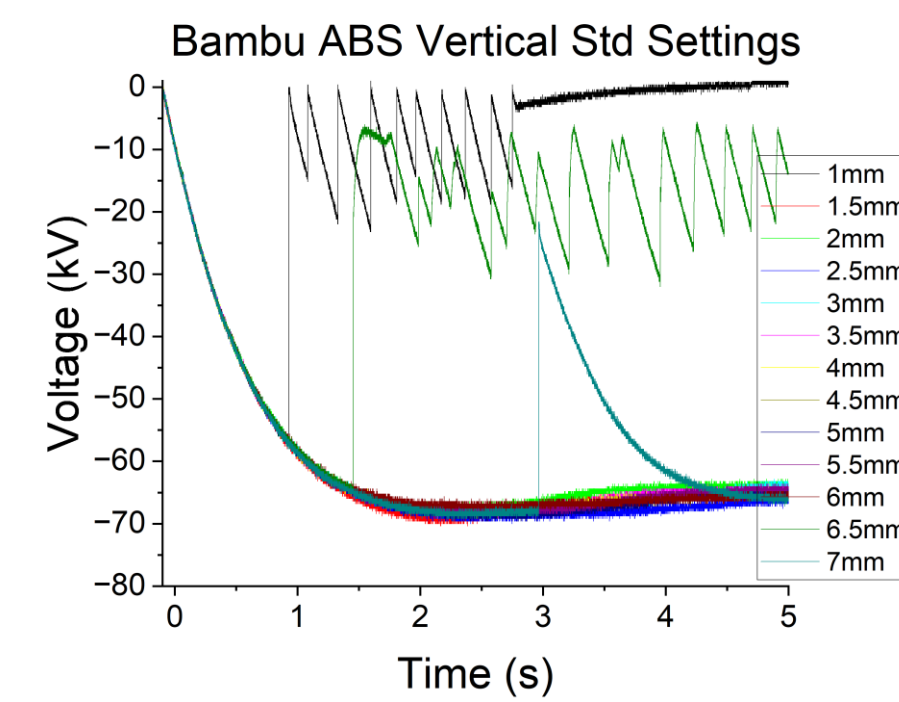
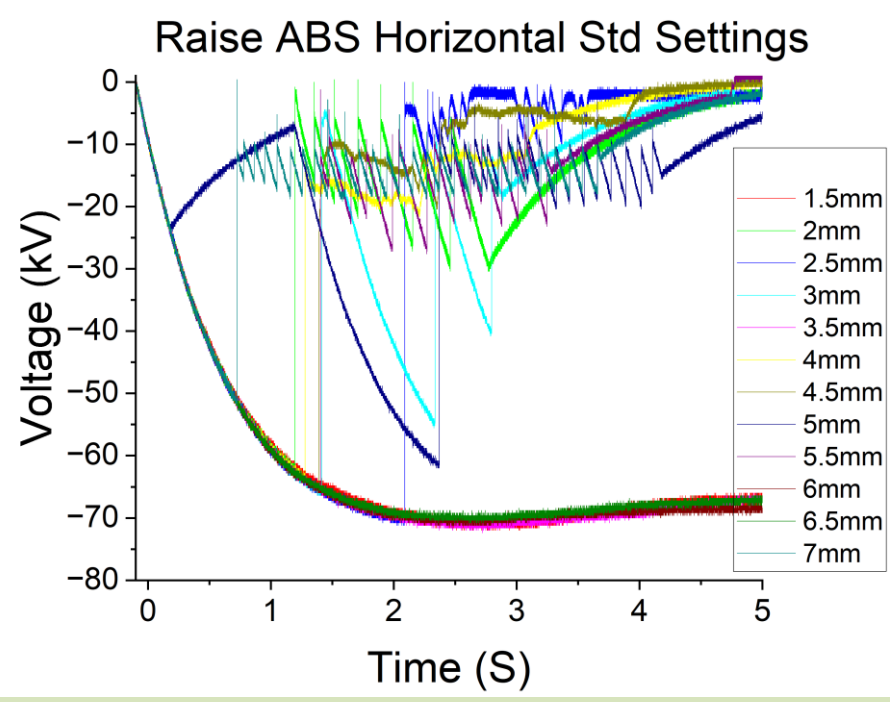
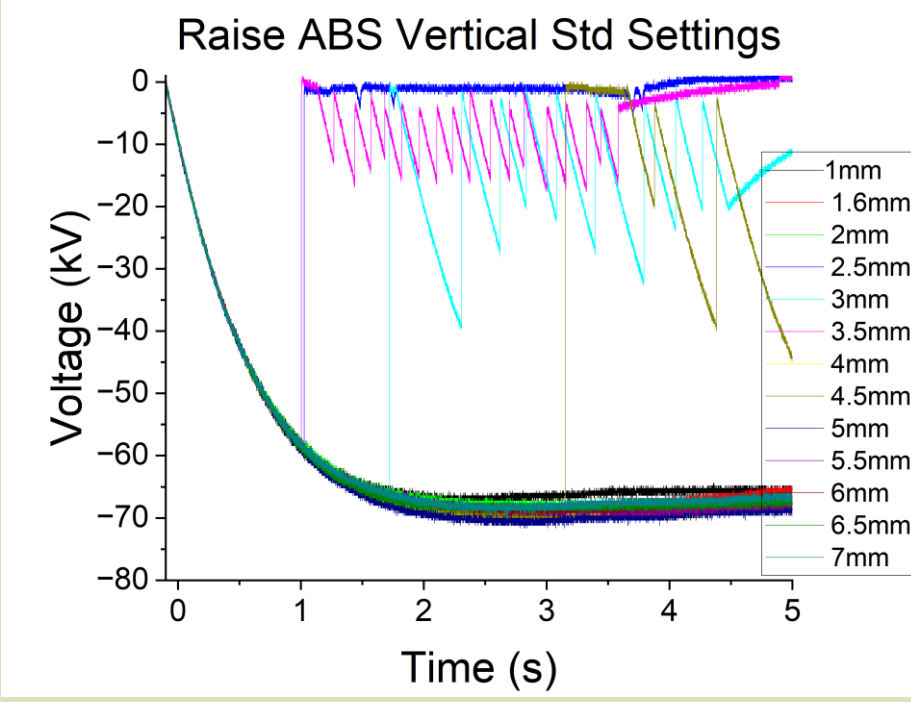
‘DC Hold off’ experiments

Tests were performed in Diala oil to prevent arcing and a small capacitance was added in parallel to the sample to ensure if failure occurred the discharge was easily visible (not enough energy in cable itself).

A Northstar probe connected to Tektronix oscilloscope recorded voltage data. The Glassman power supply was set to raise voltage to 70kV over ~1.5 seconds. To pass the test the sample must hold off voltage for > 3s.



Tests with ABS – Printer vs. Printer

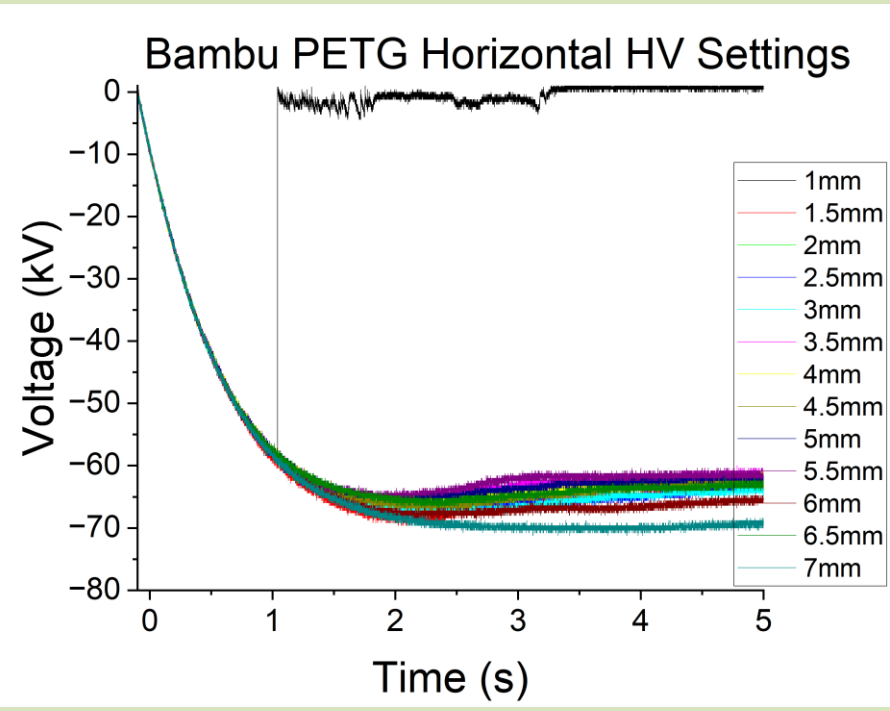
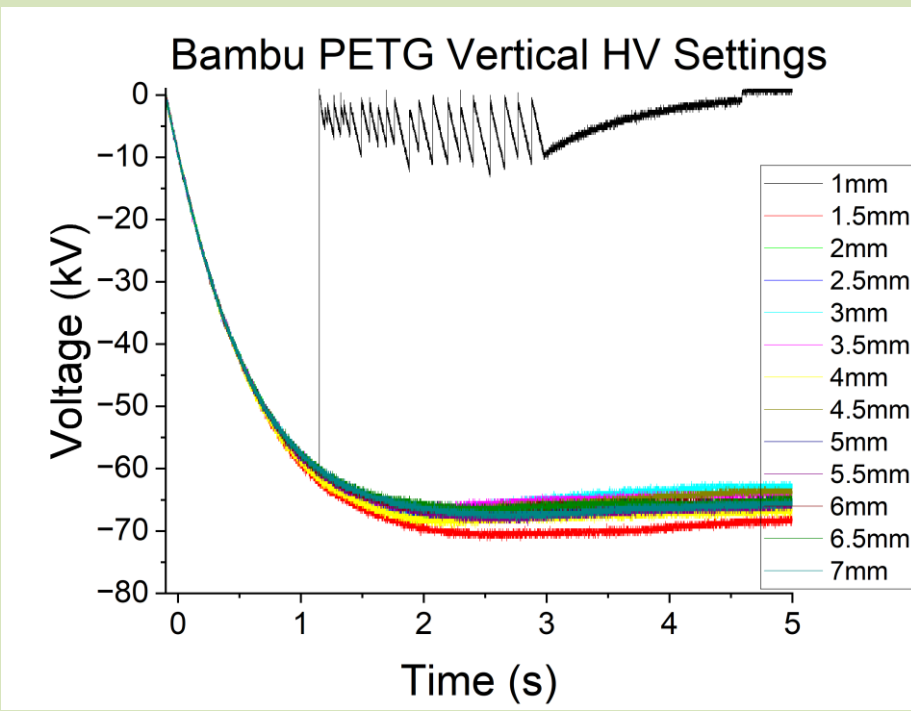
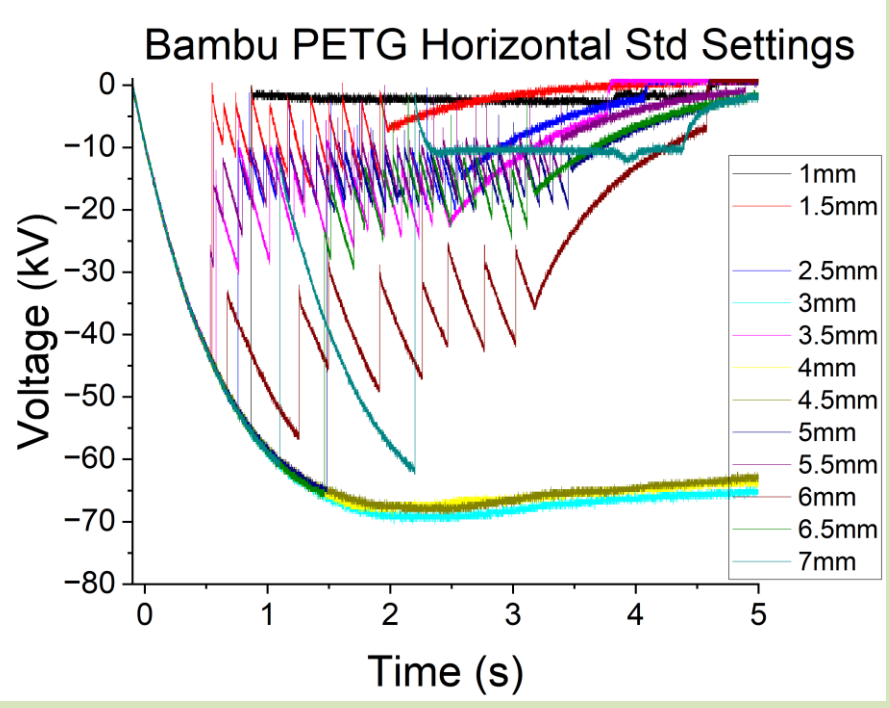
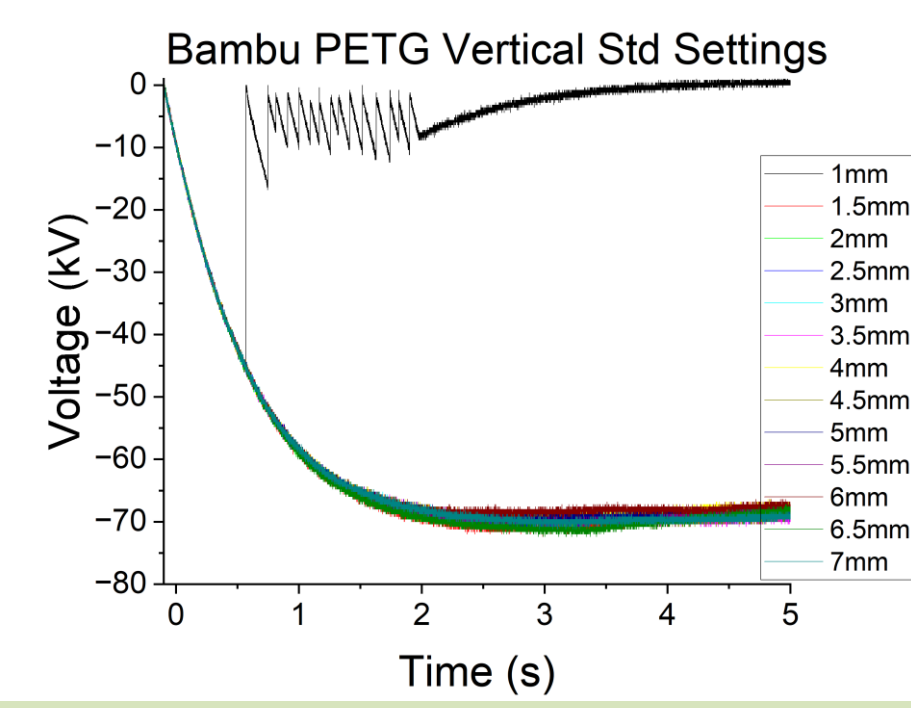


As expected (and as found by French, Hoff etc) horizontal prints do not hold off voltage as well as vertical prints

⇒ But with HV settings marked improvement in reliability of both printers in both print directions

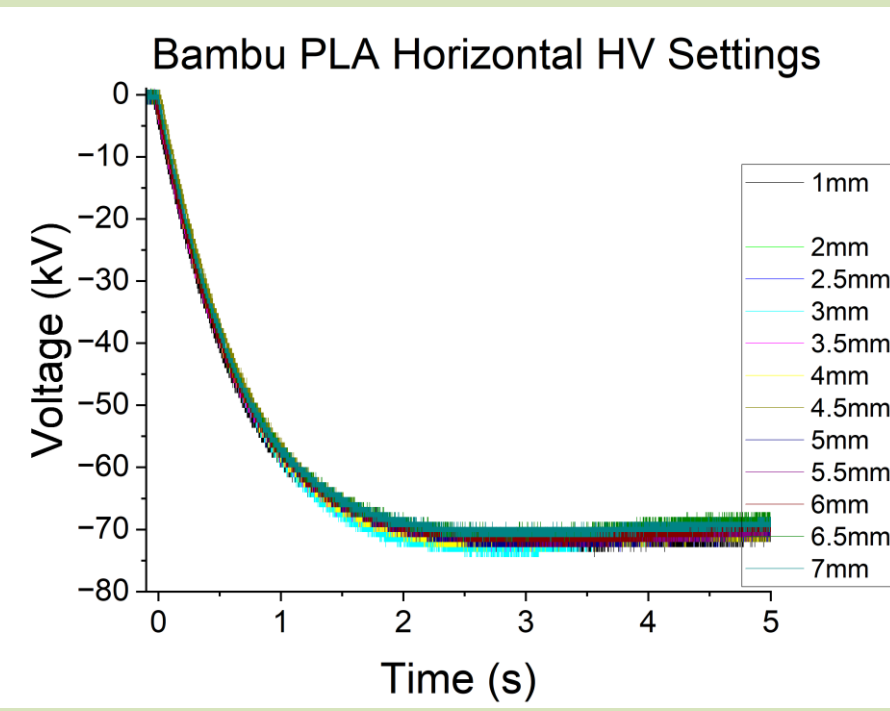
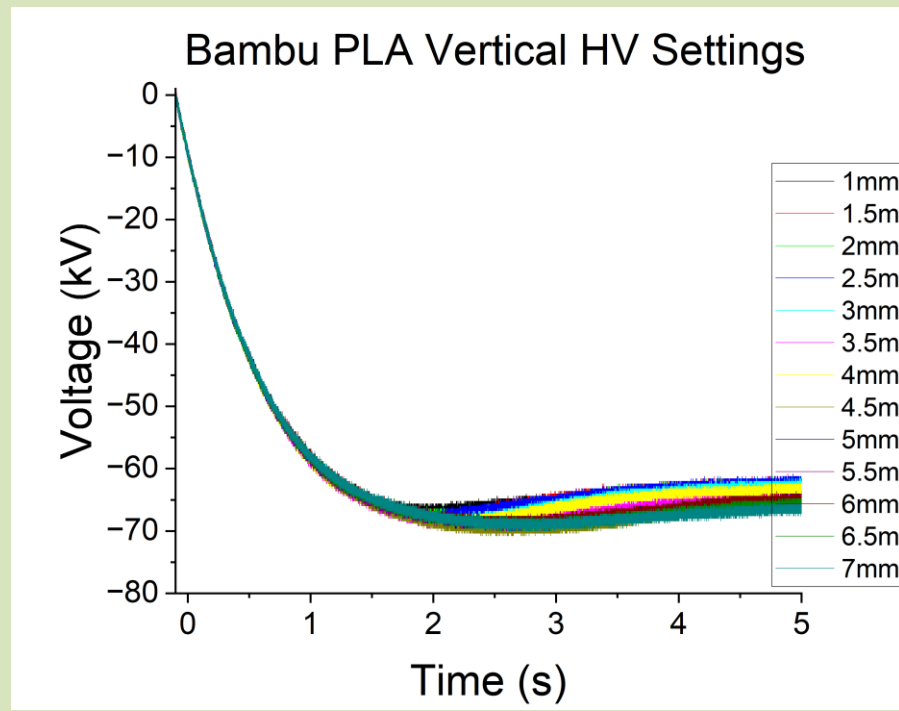
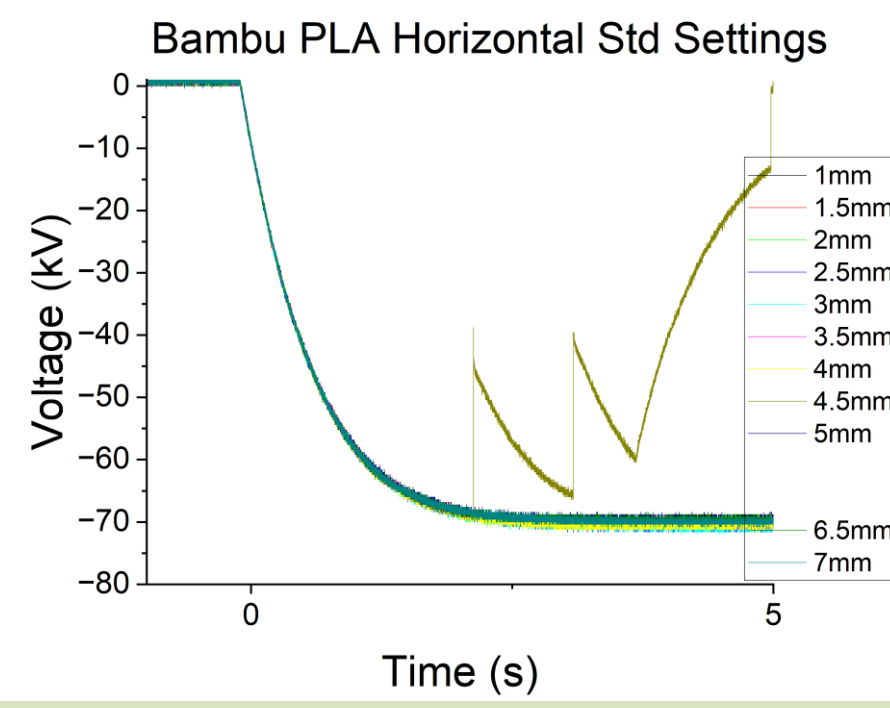
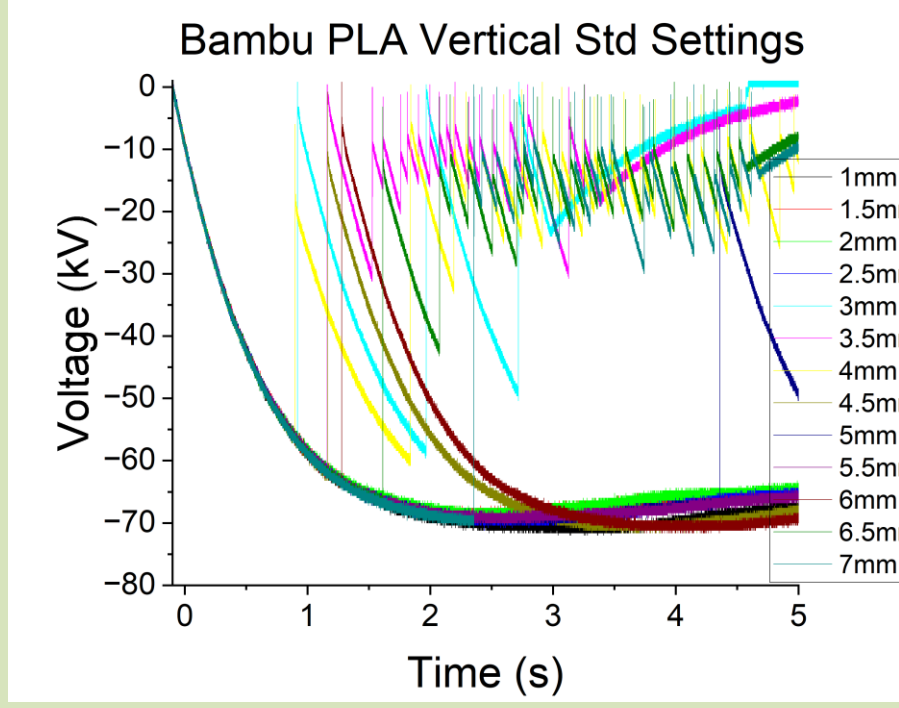
⇒ With Bambu printer even 1mm samples hold off 70kV! This is comparable to cast ABS block samples, milled to same sample size

Tests with Pet-G on Bambu Printer



⇒ With HV setting all but thinnest (1mm) samples hold 70kV

Tests with PLA on Bambu Printer

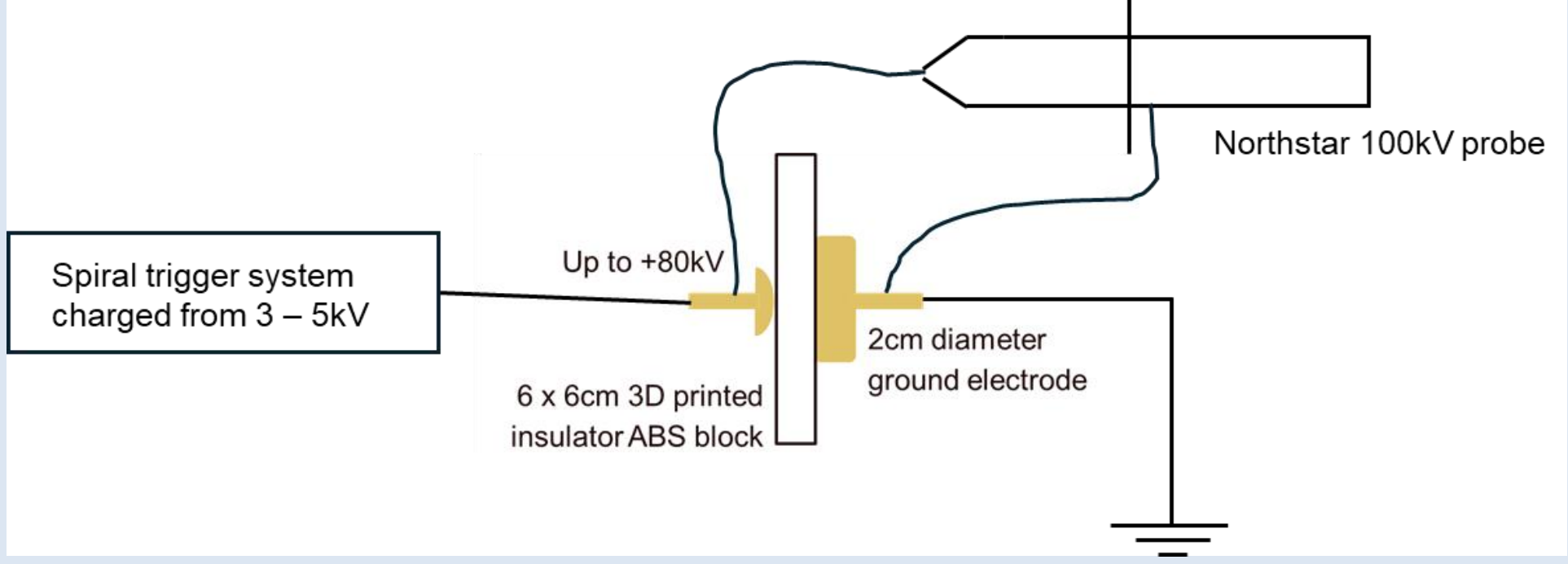
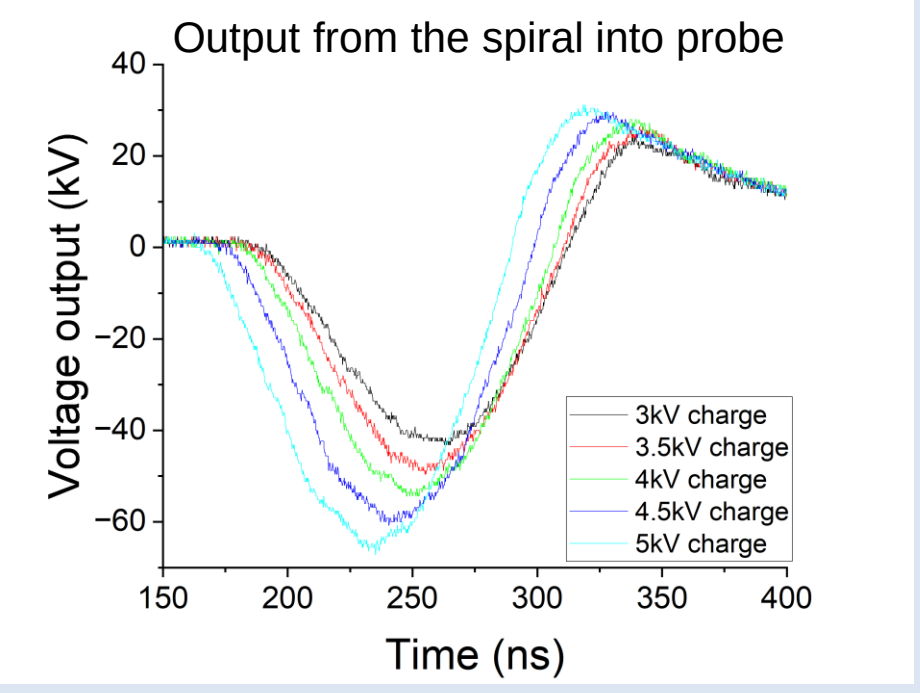


⇒ With HV setting all samples hold 70kV – strangely though with std settings, horizontal prints appear more reliable than vertical in these experiments.

Pulse experiments

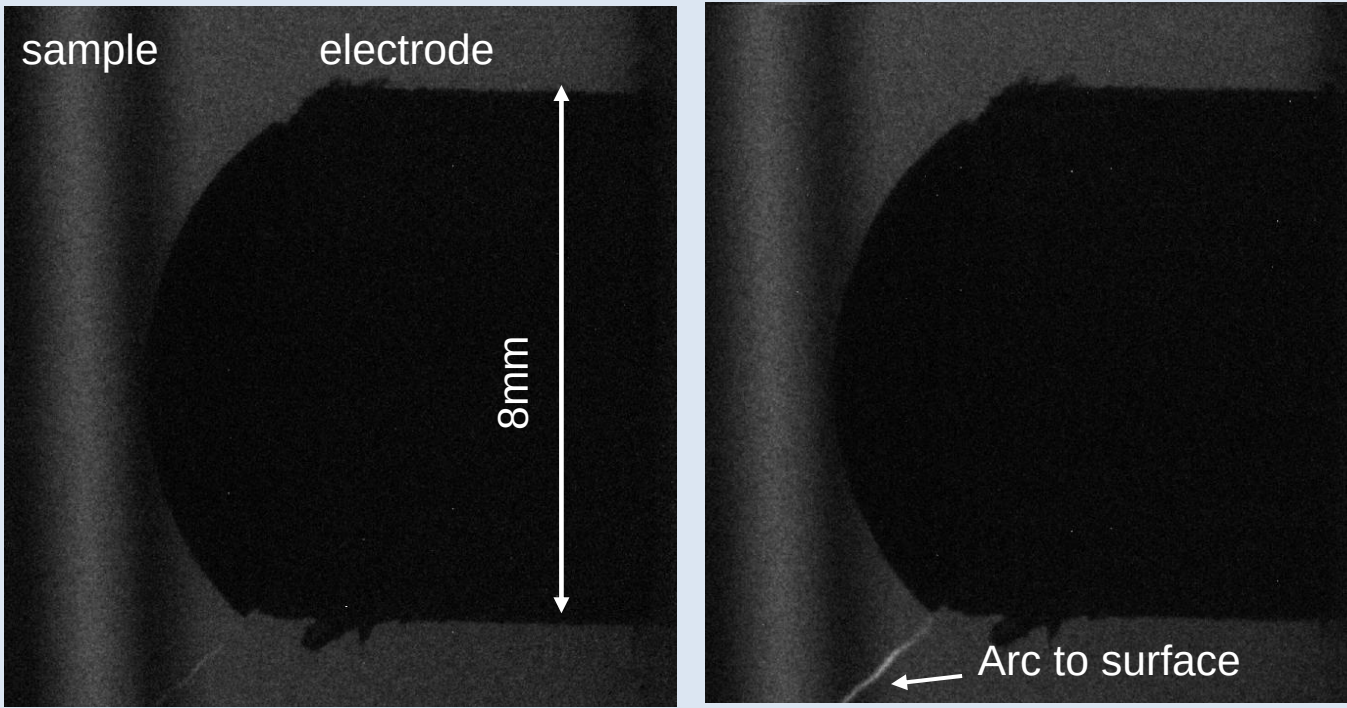
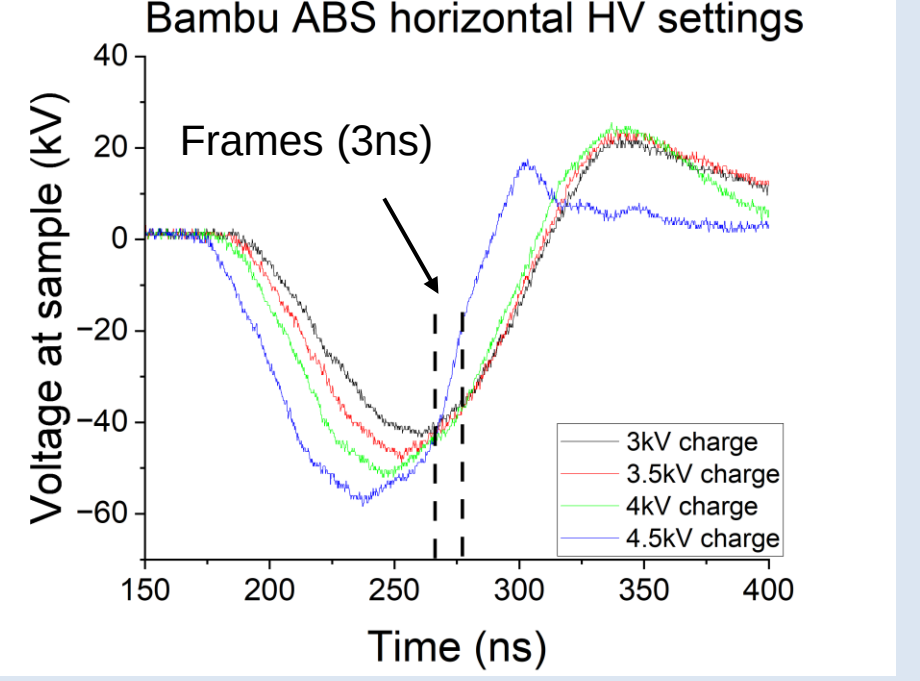
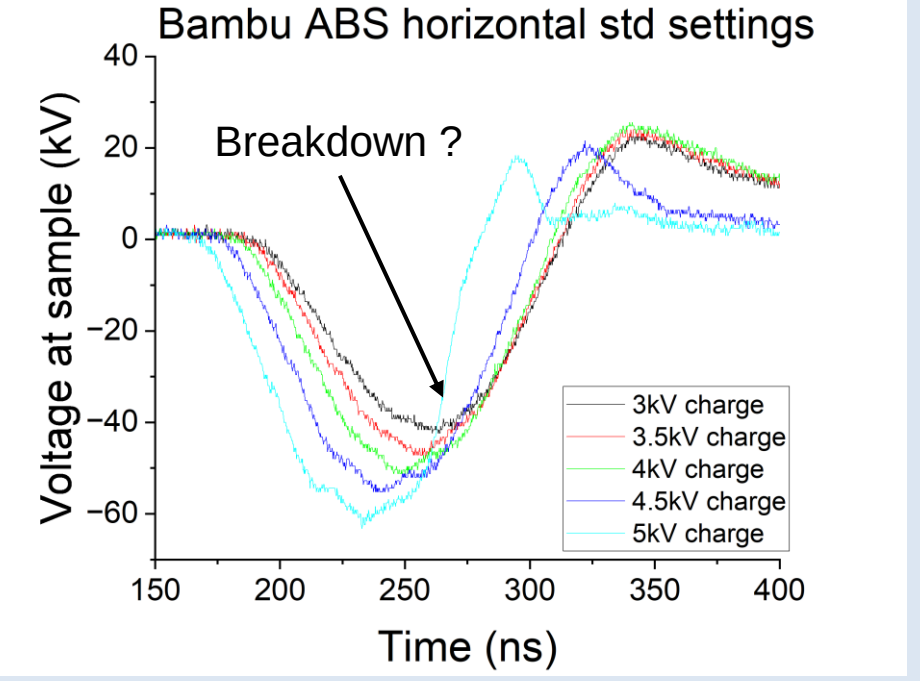
A spiral generator (24turns, solidtron switched) was utilized to provide up to 70kV at the sample with ~ 50ns risetime.

The voltage at the sample was again monitored with a Northstar probe, and a Simd 32 frame camera from Specialised Imaging observing optical emission - this helped ensure any breakdowns inferred from the voltage signal were not due to arcing around the side of the sample.



The voltage waveforms recorded suggested samples made with both standard and HV settings suffered failures when subjected to ~60kV pulses from the spiral.

However the high speed camera (and after shot inspection of samples) suggested this was actually arcing around sample itself.



Use 3D printer to our advantage! – increase path length without affecting target thickness added a series of 2mm rings around central area

⇒ no failures shown on waveform or arcs seen on camera

